XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

COURSE CONTENT AND SYLLABUS

1. Course Name	Electromagnetic Field Theory and Waves
2. Course Code	EEE302
3. Credit Value	4
4. Academic Session	2023/04
5. Lecturer(s)	Associate Prof. Dr. Zulkifly Abbas
6. Tutor(s) (if any)	
7. Course Learning Outcomes (CLO)	At the end of the course, students will be able to: 1. Describe the physical concepts of static electric fields and static magnetic fields. (C2-PLO1) 2. Apply electrostatic and magnetostatic equation to solve time-varying fields and electromagnetic wave propagation.(C3-PLO2) 3. Measure the characteristic of transmission lines using instruments and software or other specified tools (P4-PLO5)
8. Sequence of planned lessons	**Topics**
Week 1	1. Electrostatics 1.1 Coulomb's Law, Electric Field Distributions 1.2 Gauss law's
Week 2	1. Electrostatics 1.3 Energy Density in electrostatic Flux lines 1.4 Electric Fields in material space- convention and conduction currents.
Week 3	1. Electrostatics 1.5 Bio Savart Law 1.6 Magnetic Flux Density
Week 4	2. Basic Principles of Electromagnetic Wave Theory 2.1 Maxwell’s equations
Week 5	2. Basic Principles of Electromagnetic Wave Theory 2.2Time-varying fields
Week 6	2. Basic Principles of Electromagnetic WaveTheory 2.3 Plane electromagnetic wave
Week 7	2. Basic Principles of Electromagnetic WaveTheory 2.4 Plane wave reflection and dispersion 2.5 Boundary conditions
Week 8	3. EM wave propagation 3.1 Wave Equations for Conducting and Perfect Dielectric Media
Week 9	3. EM wave propagation 3.2 Relations between E and H 3.3 Wave Propagation in Lossless and Conducting Media
Week 10	3. EM wave propagation 3.4 Conductors and Dielectrics – Characterization

Week 11	3. EM wave propagation 3.5 Polarization
Week 12	4.Waveguide 4.1 Transmission Line Equations 4.2 Impedance & Smith Chart
Week 13	4.Waveguide 4.3TE mode & TM mode 4.4 Coaxial line
Week 14	4.Waveguide 4.5 VSWR
Week 15	4.Waveguide 4.6 UHF Lines as Circuit Elements

9. Required Materials (including equipment & technology)

No.	List of Practical Activities
1.	**Activities:** Describe industrial applications of waveguide.
2.	**Laboratory:** Students setup and measure the VSWR of a load, short, open circuit and horn antennas of various dimensions using Microwave Trainer Mod. MW-E/EV. (Elettronica Veneta S.P.A)
3.	**Materials/ Software used:** Use any Smith Chart simulation program(example QuickSmith) to reproduce Figure 1(lecture note 13) and determine the required inductance for an antenna with an input impedance of 50 –j40 Ohms, a 50 Ohm transmission line, and a transmitter with a desired output impedance of 50 + j0 Ohms.

10. Assessments

Continuous Assessment	**50%**
Assignments/Lab	10
Lab Report	20
Quiz	20
Final Assessment	**50%**
• Final Examination	50%

11. Assessments that align to Course Learning Outcomes (CLO)

Assessment	CLO 1	CLO2	CLO3	CLO4	CLO5
Continuous Assessment					
• Assignment	√	√	√		
• Test	√	√			
Final Assessment					
• Final Examination	√	√			

12. Main Reference(s)	1. Sadiku,M. & Nelatury,S. (2021), Element of Electromagnetics (7th Edition), Oxford: Oxford University Press 2. Hayt, W. & Buck, J. (2019) Engineering Electromagnetics (9th Edition). New York: McGrawhill Education 3. Ulaby, F. & Ravaioli, U. (2020) Fundamentals of Applied Electromagnetics (8th Edition). Essex: Pearson Education Ltd.
13. Additional Reference(s)	1. Cheng,D.K (2013), Field and Wave Electromagnetics, Essex: Pearson Education Ltd. 2. Pozar, D. (2011), Microwave Engineering, (4th edition), New York: John Wiley & Sons